

5 Quadratic Programming: Theory and Algorithms

5.1 Quadratic Programming

A *quadratic program* is an optimization problem whose objective is to minimize or maximize a quadratic function subject to a finite set of linear equality and inequality constraints. By flipping signs if necessary, a quadratic program can be written in the generic form:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2}\mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{D}\mathbf{x} \geq \mathbf{d} \end{aligned} \tag{5.1}$$

for some vectors and matrices $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{d} \in \mathbb{R}^p$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{D} \in \mathbb{R}^{p \times n}$, $\mathbf{Q} \in \mathbb{R}^{n \times n}$. As observed in Chapter 1 we may assume that $\mathbf{Q}$ is a symmetric matrix. The term *quadratic programming model* is also used to refer to a quadratic program. We will use these terms interchangeably throughout the book.

Quadratic programming models arise in a variety of practical contexts. The seminal *mean–variance model* of Markowitz and most of its variants for portfolio selection are quadratic programs as we illustrate in Example 5.1 below and discuss in full detail in Chapter 6. The popular ordinary least-squares and lasso estimation procedures in linear regression are also quadratic programs. Quadratic programs are also often solved as subproblems in the solution of more general nonlinear optimization problems.

Observe that the constraint set in (5.1) is convex since it is a system of linear inequalities. Furthermore, the objective function of (5.1) is convex when $\mathbf{Q}$ is a positive semidefinite matrix. Throughout this chapter we assume that $\mathbf{Q}$ is symmetric and positive semidefinite. Therefore problem (5.1) is a convex program.

A quadratic programming model is in *standard form* if it is written as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2}\mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned} \tag{5.2}$$

Example 5.1 (Asset allocation) Assume the one-year returns of the asset classes large stocks, small stocks, and bonds have the following correlations and standard deviations:

	Large	Small	Bonds	Standard deviation
Large	1	0.6	0.2	0.12
Small	0.6	1	0.5	0.20
Bonds	0.2	0.5	1	0.05

Determine the asset allocation of minimum risk, that is, find a portfolio comprised of these three asset classes whose return has the lowest standard deviation. Assume the portfolio can only hold long positions in each of the asset classes.

This problem can be formulated as a quadratic programming model. To that end, first construct the *covariance matrix* $\mathbf{V}$ of asset returns: this is the matrix whose (i, j) entry is the covariance of asset i and asset j ; that is, $\rho_{ij} \cdot \sigma_i \cdot \sigma_j$. Using matrix notation and ‘ $\circ$ ’ to denote the componentwise product of matrices, the covariance matrix can be computed as

$$\begin{aligned}
\mathbf{V} &= \begin{bmatrix} 1 & 0.6 & 0.2 \\ 0.6 & 1 & 0.5 \\ 0.2 & 0.5 & 1 \end{bmatrix} \circ \begin{bmatrix} 0.12 \\ 0.20 \\ 0.05 \end{bmatrix} \begin{bmatrix} 0.12 & 0.20 & 0.05 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0.6 & 0.2 \\ 0.6 & 1 & 0.5 \\ 0.2 & 0.5 & 1 \end{bmatrix} \circ \begin{bmatrix} 0.0144 & 0.024 & 0.006 \\ 0.024 & 0.04 & 0.01 \\ 0.006 & 0.01 & 0.0025 \end{bmatrix} \\
&= \begin{bmatrix} 0.0144 & 0.0144 & 0.0012 \\ 0.0144 & 0.04 & 0.005 \\ 0.0012 & 0.005 & 0.0025 \end{bmatrix}.
\end{aligned}$$

We are now ready to describe the quadratic programming formulation for the above asset allocation problem. (A more detailed discussion is given in Chapter 6.)

Quadratic programming model for asset allocation

Variables:

x_i : percentage of the portfolio invested in asset i for $i = 1, 2, 3$.

Objective (minimize the variance of the portfolio return):

$$\begin{aligned}
\min_{\mathbf{x}} \mathbf{x}^T \mathbf{V} \mathbf{x} &= \min_{x_1, x_2, x_3} (0.0144x_1^2 + 0.04x_2^2 + 0.0025x_3^2 + 0.0288x_1x_2 \\
&\quad + 0.0024x_1x_3 + 0.01x_2x_3)
\end{aligned}$$

Constraints:

$$\begin{aligned}
x_1 + x_2 + x_3 &= 1 \quad (\text{percentages add up to one}) \\
x_1, x_2, x_3 &\geq 0 \quad (\text{long-only positions}).
\end{aligned}$$

Observe that even in this small example the quadratic objective is much more concise and easier to write using matrix notation.

We now discuss the special case of a convex quadratic program without constraints. As Example 5.3 below illustrates, this kind of model arises naturally in the ordinary least-squares procedure.

Consider a quadratic program without constraints:

$$\min_{\mathbf{x}} \quad \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}. \quad (5.3)$$

The optimality conditions in this case are as follows.

Theorem 5.2 *Let $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and assume $\mathbf{Q}$ is symmetric and positive semidefinite. If (5.3) is bounded, then it attains its minimum. Furthermore, a point $\mathbf{x} \in \mathbb{R}^n$ is an optimal solution to (5.3) if and only if*

$$\mathbf{Q} \mathbf{x} + \mathbf{c} = \mathbf{0}. \quad (5.4)$$

When $\mathbf{Q}$ is positive definite, the problem (5.3) has the unique minimizer $\mathbf{x} = -\mathbf{Q}^{-1} \mathbf{c}$. When $\mathbf{Q}$ is positive semidefinite but not positive definite, the matrix $\mathbf{Q}$ is singular and the problem (5.3) is either unbounded or has multiple solutions.

Example 5.3 (Ordinary least squares) Assume $(\mathbf{x}_i, y_i)$, for $i = 1, \dots, N$, is a random sample drawn from the joint distribution of X, Y where X, Y are respectively $\mathbb{R}^p$ -valued and $\mathbb{R}$ -valued random variables. Using the *training data* $(\mathbf{x}_i, y_i)$, with $i = 1, \dots, N$, estimate a vector of coefficients $\boldsymbol{\beta}$ for the linear model

$$Y = \boldsymbol{\beta}^\top X + \epsilon.$$

The most popular approach to this problem is to find the estimate of $\boldsymbol{\beta}$ that solves the following least-squares problem:

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^N (\boldsymbol{\beta}^\top \mathbf{x}_i - y_i)^2.$$

Observe that

$$\sum_{i=1}^N (\boldsymbol{\beta}^\top \mathbf{x}_i - y_i)^2 = (\mathbf{X} \boldsymbol{\beta} - \mathbf{y})^\top (\mathbf{X} \boldsymbol{\beta} - \mathbf{y}) = \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} - 2 \mathbf{y}^\top \mathbf{X} \boldsymbol{\beta} + \mathbf{y}^\top \mathbf{y}$$

for

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}.$$

Hence the least-squares problem can be formulated as follows.

Quadratic programming formulation for least-squares estimation.

Variables:

$\boldsymbol{\beta}$: vector of coefficients in the linear model $Y = \boldsymbol{\beta}^\top X + \epsilon$.

Objective:

$$\min_{\boldsymbol{\beta}} \quad \frac{1}{2} \boldsymbol{\beta}^\top \mathbf{Q} \boldsymbol{\beta} - \mathbf{b}^\top \boldsymbol{\beta},$$

where $\mathbf{Q} := \mathbf{X}^\top \mathbf{X}$, $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$.

Constraints: None.

By applying Theorem 5.2 we obtain the widely known solution to the least-squares problem:

$$\hat{\boldsymbol{\beta}} := \mathbf{Q}^{-1}\mathbf{b} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y},$$

provided the $N \times p$ matrix $\mathbf{X}$ has full column rank. This latter condition usually holds in the typical practical situation when there are more observations than predictor variables; that is, when $N > p$. However, the case $N < p$ occurs as well. In this kind of situation the matrix $\mathbf{X}$ is never full column rank so the ordinary least-squares approach is not appropriate. Section 5.6.2 describes two popular variants for this kind of situation, namely *ridge regression* and *lasso regression*, both of which can be seen as modifications of the ordinary least-squares procedure.

5.2 Numerical Quadratic Programming Solvers

As with linear programming, there are a variety of highly efficient, fast, and reliable commercial and open-source software packages for convex quadratic programming. Most of these packages implement versions of the algorithms sketched in Section 5.5 below. We illustrate two of these solvers by applying them to Example 5.1.

Excel Solver

Figure 5.1 displays a printout of an Excel spreadsheet implementation of the quadratic programming model for Example 5.1 as well as the dialog box obtained when we run the Excel add-in **Solver**. The spreadsheet model contains the three components of the quadratic program. The decision variables are in the range B20:D20. The objective function is in cell E22. The Excel formula in this cell, using matrix operations, is as follows:

$$\text{MMULT}(B20 : D20, \text{MMULT}(B16 : D18, \text{TRANSPOSE}(B20 : D20))).$$

The left-hand and right-hand sides of the equality constraint are in cells E20 and G20 respectively.

MATLAB CVX

Figure 5.2 displays a CVX script for the same problem. The script can be run provided the freely available CVX toolbox is installed.

Using either of these solvers we obtain the optimal solution to the problem in Example 5.1:

$$\mathbf{x}^* = \begin{bmatrix} 0.0897 \\ 0 \\ 0.9103 \end{bmatrix}.$$

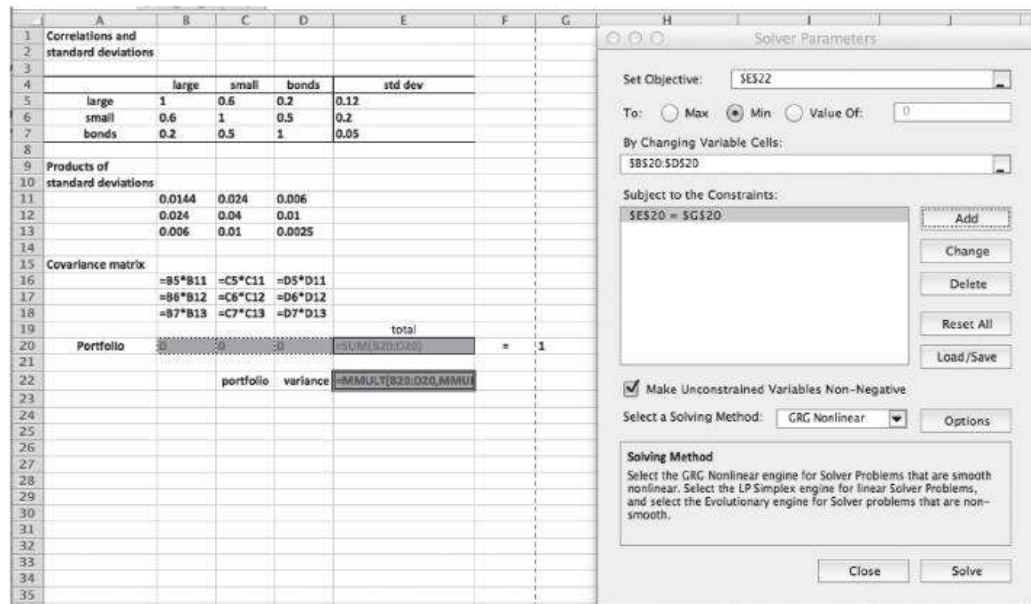


Figure 5.1 Spreadsheet implementation and the Solver dialog box for the asset allocation model

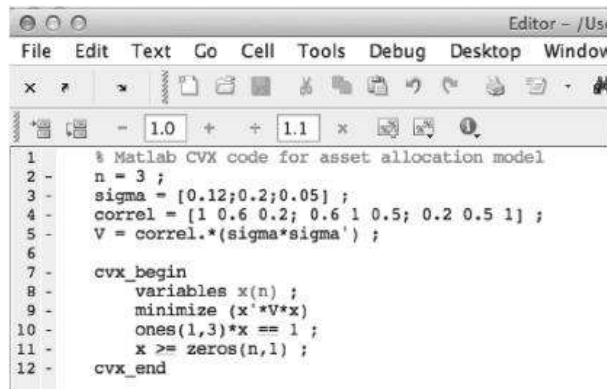


Figure 5.2 MATLAB CVX code for the asset allocation model

5.3 Sensitivity Analysis

As is the case for linear programming, the process of solving a quadratic program also generates some interesting *sensitivity information* via the so-called *Lagrange multipliers* associated with the constraints. Assume the constraints of a quadratic program, and hence the Lagrange multipliers, are indexed by $i = 1, \dots, m$. The *Lagrange multiplier* y_i^* of the i th constraint has the following sensitivity interpretation:

If the right-hand side of the i th constraint changes by Δ , then the optimal value of the quadratic program changes by approximately $\Delta \cdot y_i^*$ for small Δ .

Unlike the shadow prices of a linear program, the Lagrange multipliers only give an approximation of the change in the optimal objective value. The situation is akin to how the derivative of a quadratic (or more general nonlinear) function at a particular point gives an approximation of the change in the function value when that point changes.

Both Excel Solver and MATLAB CVX compute the Lagrange multipliers implicitly. To make this information explicit in Excel Solver, we request a sensitivity report after running Solver as shown in Figure 5.3.

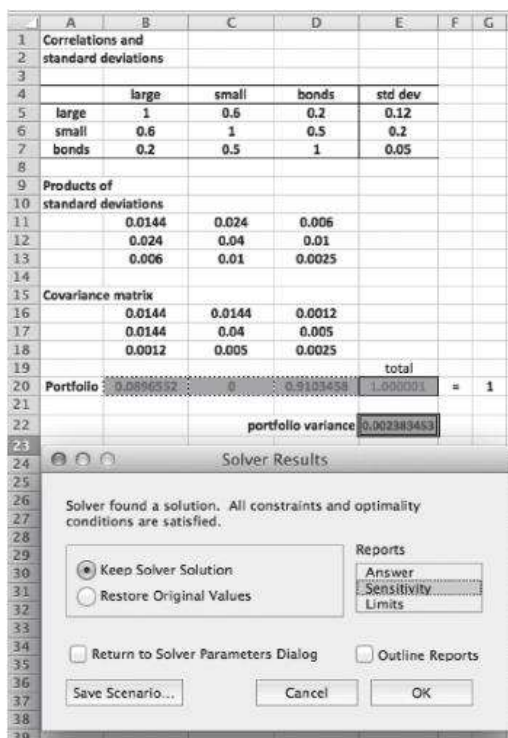


Figure 5.3 Requesting sensitivity report in Solver

Figure 5.4 displays the sensitivity report for Example 5.1. The values y_i^* can be found in the column labeled “Lagrange Multiplier”. In CVX this information can also be obtained by including a line of code to save the dual information y as shown in Figure 5.5. Both solvers yield the dual value $y^* = 0.0047669$.

5.4 *Duality and Optimality Conditions

As in linear programming, there is a *dual* quadratic program associated with every *primal* quadratic programming problem, and this dual can be obtained via the Lagrangian function. Throughout this section consider the *primal* quadratic

Variable Cells

Cell	Name	Final Value	Reduced Gradient
\$B\$20	Portfolio large	0.089655197	0
\$C\$20	Portfolio small	0	0.006918652
\$D\$20	Portfolio bonds	0.910345803	0

Constraints

Cell	Name	Final Value	Lagrange Multiplier
\$E\$20	Portfolio total	1.000001	0.004766915

Figure 5.4 Sensitivity report

```

1 % Matlab CVX code for asset allocation model
2 % Request dual variables
3 n = 3 ;
4 sigma = [0.12;0.2;0.05] ;
5 correl = [1 0.6 0.2; 0.6 1 0.5; 0.2 0.5 1] ;
6 V = correl.*(sigma*sigma') ;
7
8 cvx_begin
9     variables x(n) ;
10    dual variable y ;
11    minimize (x'*V*x)
12    y: ones(1,3)*x == 1 ;
13    x >= zeros(n,1) ;
14 cvx_end

```

Figure 5.5 MATLAB CVX code with dual variables

program

$$\begin{aligned}
 \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} \\
 \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \\
 & \mathbf{D} \mathbf{x} \geq \mathbf{d},
 \end{aligned} \tag{5.5}$$

where $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{Q} \in \mathbb{R}^{n \times n}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{D} \in \mathbb{R}^{p \times n}$, $\mathbf{d} \in \mathbb{R}^p$, and $\mathbf{Q}$ is symmetric and positive semidefinite.

The *Lagrangian function* associated with (5.5) is

$$L(\mathbf{x}, \mathbf{y}, \mathbf{s}) := \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + \mathbf{y}^T (\mathbf{b} - \mathbf{A} \mathbf{x}) + \mathbf{s}^T (\mathbf{d} - \mathbf{D} \mathbf{x}).$$

The constraints of (5.5) can be encoded via the Lagrangian function through the following observation: For a given vector $\mathbf{x}$

$$\max_{\substack{\mathbf{y}, \mathbf{s} \\ \mathbf{s} \geq \mathbf{0}}} L(\mathbf{x}, \mathbf{y}, \mathbf{s}) = \begin{cases} \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} & \text{if } \mathbf{A} \mathbf{x} = \mathbf{b} \text{ and } \mathbf{D} \mathbf{x} \geq \mathbf{d} \\ +\infty & \text{otherwise.} \end{cases}$$

Therefore the primal problem (5.5) can be written as

$$\min_{\mathbf{x}} \max_{\substack{\mathbf{y}, \mathbf{s} \\ \mathbf{s} \geq \mathbf{0}}} L(\mathbf{x}, \mathbf{y}, \mathbf{s}).$$

The dual problem is obtained by flipping the order of the min and max operations:

$$\max_{\substack{\mathbf{y}, \mathbf{s} \\ \mathbf{s} \geq \mathbf{0}}} \min_{\mathbf{x}} L(\mathbf{x}, \mathbf{y}, \mathbf{s}).$$

It is easy to see that the dual problem can be written as follows:

$$\begin{aligned} \max_{\mathbf{x}, \mathbf{y}, \mathbf{s}} \quad & \mathbf{b}^\top \mathbf{y} + \mathbf{d}^\top \mathbf{s} - \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}^\top \mathbf{y} + \mathbf{D}^\top \mathbf{s} - \mathbf{Q} \mathbf{x} = \mathbf{c} \\ & \mathbf{s} \geq \mathbf{0}. \end{aligned} \tag{5.6}$$

In particular, when the primal problem is in standard form (5.2), the dual problem is

$$\begin{aligned} \max_{\mathbf{x}, \mathbf{y}, \mathbf{s}} \quad & \mathbf{b}^\top \mathbf{y} - \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}^\top \mathbf{y} - \mathbf{Q} \mathbf{x} + \mathbf{s} = \mathbf{c} \\ & \mathbf{s} \geq \mathbf{0}. \end{aligned}$$

Observe that the dual problem of a quadratic program is again a quadratic program. Note that, unlike the case of linear programming, some primal-like variables $\mathbf{x}$ also appear in the dual problem. As in linear programming, there is a deep connection between the primal problem (5.5) and its dual (5.6). The next result follows by construction.

Theorem 5.4 (Weak duality) *Assume $\mathbf{x}$ is a feasible point for (5.5) and $(\tilde{\mathbf{x}}, \mathbf{y}, \mathbf{s})$ is a feasible point for (5.6). Then*

$$\mathbf{b}^\top \mathbf{y} + \mathbf{d}^\top \mathbf{s} - \frac{1}{2} \tilde{\mathbf{x}}^\top \mathbf{Q} \tilde{\mathbf{x}} \leq \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}.$$

Proof If $\mathbf{x}$ and $(\tilde{\mathbf{x}}, \mathbf{y}, \mathbf{s})$ satisfy the above assumptions then

$$\begin{aligned} \mathbf{b}^\top \mathbf{y} + \mathbf{d}^\top \mathbf{s} - \frac{1}{2} \tilde{\mathbf{x}}^\top \mathbf{Q} \tilde{\mathbf{x}} &\leq (\mathbf{A} \mathbf{x})^\top \mathbf{y} + (\mathbf{D} \mathbf{x})^\top \mathbf{s} - \frac{1}{2} \tilde{\mathbf{x}}^\top \mathbf{Q} \tilde{\mathbf{x}} \\ &= (\mathbf{A}^\top \mathbf{y} + \mathbf{D}^\top \mathbf{s})^\top \mathbf{x} - \frac{1}{2} \tilde{\mathbf{x}}^\top \mathbf{Q} \tilde{\mathbf{x}} \\ &= (\mathbf{c} + \mathbf{Q} \tilde{\mathbf{x}})^\top \mathbf{x} - \frac{1}{2} \tilde{\mathbf{x}}^\top \mathbf{Q} \tilde{\mathbf{x}} \\ &= \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} - \frac{1}{2} (\mathbf{x} - \tilde{\mathbf{x}})^\top \mathbf{Q} (\mathbf{x} - \tilde{\mathbf{x}}) \\ &\leq \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}. \end{aligned}$$

□

The following much deeper result also holds.

Theorem 5.5 (Strong duality) *Assume one of the problems (5.5) or (5.6) is feasible. Then this problem is bounded if and only if the other one is feasible. In that case both problems have optimal solutions and their optimal values are the same.*

We refer the reader to Güler (2010) or Nocedal and Wright (2006) for a proof of Theorem 5.5. This result is closely tied to certain kinds of *separation theorems* for convex sets. For details see Güler (2010, chapters 6 and 11). A powerful consequence of Theorem 5.5 is the following characterization of the solutions to both (5.5) and (5.6).

Theorem 5.6 (Optimality conditions) *The vectors $\mathbf{x} \in \mathbb{R}^n$ and $(\tilde{\mathbf{x}}, \mathbf{y}, \mathbf{s}) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p$ are optimal solutions to (5.5) and (5.6) respectively if and only if $\mathbf{Q}\mathbf{x} = \mathbf{Q}\tilde{\mathbf{x}}$ and*

$$\begin{aligned} \mathbf{Q}\mathbf{x} + \mathbf{c} - \mathbf{A}^\top \mathbf{y} - \mathbf{D}^\top \mathbf{s} &= \mathbf{0} \\ \mathbf{A}\mathbf{x} - \mathbf{b} &= \mathbf{0} \\ \mathbf{D}\mathbf{x} - \mathbf{d} &\geq \mathbf{0} \\ \mathbf{s} &\geq \mathbf{0} \\ (\mathbf{D}\mathbf{x} - \mathbf{d})_i s_i &= 0, \quad i = 1, \dots, p. \end{aligned} \tag{5.7}$$

For a quadratic program in standard form (5.2), the optimality conditions (5.7) can be written as follows:

$$\begin{aligned} -\mathbf{Q}\mathbf{x} + \mathbf{A}^\top \mathbf{y} + \mathbf{s} &= \mathbf{c} \\ \mathbf{A}\mathbf{x} &= \mathbf{b} \\ \mathbf{x} &\geq \mathbf{0} \\ \mathbf{s} &\geq \mathbf{0} \\ x_i s_i &= 0, \quad i = 1, \dots, n. \end{aligned} \tag{5.8}$$

Observe that (5.8) nicely extends the optimality conditions (2.9) for linear programming in standard form.

The optimality conditions (5.7) can be seen as “saddle-point” conditions for the Lagrangian function

$$L(\mathbf{x}, \mathbf{y}, \mathbf{s}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x} + \mathbf{y}^\top (\mathbf{b} - \mathbf{A}\mathbf{x}) + \mathbf{s}^\top (\mathbf{d} - \mathbf{D}\mathbf{x}).$$

We next discuss the special case of a quadratic program with equality constraints only. Consider the problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b}, \end{aligned} \tag{5.9}$$

where $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{Q} \in \mathbb{R}^{n \times n}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{Q}$ is symmetric and positive semidefinite. In this case the optimality conditions (5.7) simplify to

$$\begin{aligned} \mathbf{Q}\mathbf{x} + \mathbf{c} - \mathbf{A}^\top \mathbf{y} &= \mathbf{0} \\ \mathbf{A}\mathbf{x} - \mathbf{b} &= \mathbf{0}. \end{aligned} \tag{5.10}$$

The optimality conditions (5.10) in turn can be stated in terms of the Lagrangian function of (5.9):

$$L(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} + \mathbf{y}^\top (\mathbf{b} - \mathbf{A} \mathbf{x}).$$

Indeed observe that (5.10) can be succinctly written as

$$\nabla L(\mathbf{x}, \mathbf{y}) = \mathbf{0}.$$

When $\mathbf{Q}$ is positive definite and $\mathbf{A}$ has full row rank, problem (5.9) has a unique minimizer $\mathbf{x}$ and a unique Lagrange multiplier $\mathbf{y}$ given by

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{Q} & -\mathbf{A}^\top \\ \mathbf{A} & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} -\mathbf{c} \\ \mathbf{b} \end{bmatrix}.$$

In particular, if $\mathbf{Q}$ is positive definite and $\mathbf{A}$ has full row rank, then the minimizer and vector of Lagrange multipliers for the problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned} \tag{5.11}$$

are respectively

$$\begin{aligned} \mathbf{x}^* &= \mathbf{Q}^{-1} \mathbf{A}^\top (\mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^\top)^{-1} \mathbf{b} \\ \mathbf{y}^* &= (\mathbf{A} \mathbf{Q}^{-1} \mathbf{A}^\top)^{-1} \mathbf{b}. \end{aligned}$$

Example 5.7 (Asset allocation) Consider the same problem as in Example 5.1 but assume this time that the portfolio is allowed to hold short positions.

The formulation for this modification of Example 5.1 is straightforward: just drop the non-negativity constraint on the variables. Thus we obtain the quadratic programming model

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{V} \mathbf{x} \\ \text{s.t.} \quad & \mathbf{1}^\top \mathbf{x} = 1. \end{aligned}$$

From the above discussion it readily follows that the optimal solution and Lagrange multiplier are

$$\begin{aligned} \mathbf{x}^* &= \frac{1}{\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1}} \mathbf{V}^{-1} \mathbf{1} \\ y^* &= \frac{1}{\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1}}. \end{aligned}$$

For the particular value of $\mathbf{V}$ in Example 5.1 we get the following optimal solution and Lagrange multiplier

$$\mathbf{x}^* = \begin{bmatrix} 0.1934 \\ -0.1406 \\ 0.9472 \end{bmatrix}, \quad y^* = 0.001897074.$$

5.5 *Algorithms

We next sketch extensions of the two main algorithmic schemes for linear programming discussed in Chapter 2. The first scheme, namely *active-set methods*, can be seen as an analog of the simplex method. The second scheme, namely *interior-point methods*, is a straightforward extension from the linear programming to the quadratic programming context.

5.5.1 Active-Set Methods

Active-set methods are based on the following key observation. Assume $\bar{\mathbf{x}}$ is an optimal solution to (5.5) and

$$I := \{i = 1, \dots, p : (\mathbf{D}\bar{\mathbf{x}} - \mathbf{d})_i = 0\}.$$

Then the optimality conditions (5.7) can be rewritten as

$$\begin{aligned} \mathbf{Q}\mathbf{x} + \mathbf{c} - \mathbf{A}^\top \mathbf{y} - \mathbf{D}_I^\top \mathbf{s}_I &= \mathbf{0} \\ \mathbf{A}\mathbf{x} - \mathbf{b} &= \mathbf{0} \\ \mathbf{D}_I \mathbf{x} - \mathbf{d}_I &= \mathbf{0} \\ \mathbf{s}_I &\geq \mathbf{0}. \end{aligned} \tag{5.12}$$

If we ignore the last constraint $\mathbf{s}_I \geq \mathbf{0}$, the remaining conditions in (5.12) are precisely the optimality conditions of the problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \\ & \mathbf{D}_I \mathbf{x} = \mathbf{d}_I. \end{aligned} \tag{5.13}$$

This suggests an algorithmic strategy to solve (5.5): guess the *active set* I and solve the subproblem (5.13). If the solution $\bar{\mathbf{x}}$ to this subproblem satisfies the other conditions in (5.7) then stop. Otherwise, make a new guess for I . Algorithm 5.1 gives a possible version of this strategy.

Each main iteration of Algorithm 5.1 requires solving the following subproblem for some current trial solution $\bar{\mathbf{x}}$ and trial active set I :

$$\begin{aligned} \min_{\Delta \mathbf{x}} \quad & \frac{1}{2} (\Delta \mathbf{x})^\top \mathbf{Q} \Delta \mathbf{x} + (\mathbf{Q}\bar{\mathbf{x}} + \mathbf{c})^\top \Delta \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \Delta \mathbf{x} = \mathbf{0} \\ & \mathbf{D}_I \Delta \mathbf{x} = \mathbf{0}. \end{aligned} \tag{5.14}$$

To update the trial solution we also need to compute the step length

$$\alpha := \min \left\{ 1, \min_{\substack{i \notin I \\ \mathbf{D}_i \Delta \mathbf{x} < 0}} \frac{\mathbf{d}_i - \mathbf{D}_i \Delta \mathbf{x}}{\mathbf{D}_i \Delta \mathbf{x}} \right\}. \tag{5.15}$$

Algorithm 5.1 Active-set method

```

1: choose  $\mathbf{x}_0$  feasible for (5.5) and  $I_0 \subseteq \{i : \mathbf{D}_i \mathbf{x}_0 = \mathbf{d}_i, i = 1, \dots, p\}$ 
2: for  $k = 0, 1, \dots$  do
3:   solve (5.14) for  $I = I_k$  and  $\bar{\mathbf{x}} = \mathbf{x}_k$ 
4:   if  $\Delta \mathbf{x} = \mathbf{0}$  then
5:     compute the Lagrange multipliers  $\bar{\mathbf{s}}_I$  of (5.14) for  $I = I_k$  and  $\bar{\mathbf{x}} = \mathbf{x}_k$ 
6:     if  $\bar{\mathbf{s}}_I \geq \mathbf{0}$  then HALT  $\bar{\mathbf{x}}$  is an optimal solution to (5.5)
7:     else
8:       let  $j := \arg \min_{i \in I} \bar{\mathbf{s}}_i$ ,  $I_{k+1} := I_k \setminus \{j\}$ , and  $\mathbf{x}_{k+1} := \mathbf{x}_k$ 
9:     end if
10:  else
11:    compute  $\alpha$  via (5.15) for  $I = I_k$  and let  $\mathbf{x}_{k+1} := \mathbf{x}_k + \alpha \Delta \mathbf{x}$ 
12:    if  $\alpha_k$  has a blocking constraint  $j$  then  $I_{k+1} := I_k \cup \{j\}$ 
13:    else  $I_{k+1} := I_k$ 
14:    end if
15:  end if
16: end for

```

We say that the step length α computed in (5.15) has a *blocking constraint*, $j \notin I$, if

$$\alpha = \min_{\substack{i \notin I \\ \mathbf{D}_i \Delta \mathbf{x} < 0}} \frac{\mathbf{d}_i - \mathbf{D}_i \Delta \mathbf{x}}{\mathbf{D}_i \Delta \mathbf{x}} = \frac{\mathbf{d}_j - \mathbf{D}_j \Delta \mathbf{x}}{\mathbf{D}_j \Delta \mathbf{x}} < 1.$$

And we say that α has no blocking constraints when

$$\alpha = 1 < \min_{\substack{i \notin I \\ \mathbf{D}_i \Delta \mathbf{x} < 0}} \frac{\mathbf{d}_i - \mathbf{D}_i \Delta \mathbf{x}}{\mathbf{D}_i \Delta \mathbf{x}}.$$

5.5.2 Interior-Point Methods

For notational convenience and without loss of generality we assume that the problem of interest is in standard form (5.2).

As in the linear programming case (Section 2.7.3), interior-point methods generate a sequence of iterates that satisfy $\mathbf{x}, \mathbf{s} > \mathbf{0}$. Each iteration of the algorithm aims to make progress towards satisfying $-\mathbf{Q}\mathbf{x} + \mathbf{A}^\top \mathbf{y} + \mathbf{s} = \mathbf{c}$, $\mathbf{A}\mathbf{x} = \mathbf{b}$, and $x_i s_i = 0$, with $i = 1, \dots, n$.

As before we use the following notational convention: Given a vector $\mathbf{x} \in \mathbb{R}^n$, let $\mathbf{X} \in \mathbb{R}^{n \times n}$ denote the diagonal matrix defined by $X_{ii} = x_i$, with $i = 1, \dots, n$, and let $\mathbf{1} \in \mathbb{R}^n$ denote the vector whose components are all 1s. The optimality conditions (5.8) can be restated as

$$\begin{bmatrix} -\mathbf{Q}\mathbf{x} + \mathbf{A}^\top \mathbf{y} + \mathbf{s} - \mathbf{c} \\ \mathbf{A}\mathbf{x} - \mathbf{b} \\ \mathbf{X}\mathbf{S}\mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{x}, \mathbf{s} \geq \mathbf{0}.$$

Given $\mu > 0$, let $(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{s}(\mu))$ be the solution to the following perturbed version of the above optimality conditions:

$$\begin{bmatrix} -\mathbf{Q}\mathbf{x} + \mathbf{A}^\top \mathbf{y} + \mathbf{s} - \mathbf{c} \\ \mathbf{A}\mathbf{x} - \mathbf{b} \\ \mathbf{X}\mathbf{S}\mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mu \mathbf{1} \end{bmatrix}, \quad \mathbf{x}, \mathbf{s} > \mathbf{0}.$$

The first condition above can be written as $\mathbf{r}_\mu(\mathbf{x}, \mathbf{y}, \mathbf{s}) = \mathbf{0}$ for the *residual vector*

$$\mathbf{r}_\mu(\mathbf{x}, \mathbf{y}, \mathbf{s}) := \begin{bmatrix} -\mathbf{Q}\mathbf{x} + \mathbf{A}^\top \mathbf{y} + \mathbf{s} - \mathbf{c} \\ \mathbf{A}\mathbf{x} - \mathbf{b} \\ \mathbf{X}\mathbf{S}\mathbf{1} - \mu \mathbf{1} \end{bmatrix}.$$

The *central path* is the set $\{(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{s}(\mu)) : \mu > 0\}$. It is intuitively clear that $(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{s}(\mu))$ converges to an optimal solution to both (5.2) and its dual. This suggests the following algorithmic strategy. Suppose $(\mathbf{x}, \mathbf{y}, \mathbf{s})$ is “near” $(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{s}(\mu))$ for some $\mu > 0$. Use $(\mathbf{x}, \mathbf{y}, \mathbf{s})$ to move to a better point $(\mathbf{x}^+, \mathbf{y}^+, \mathbf{s}^+)$ “near” $(\mathbf{x}(\mu^+), \mathbf{y}(\mu^+), \mathbf{s}(\mu^+))$ for some $\mu^+ < \mu$.

It can be shown that if a point $(\mathbf{x}, \mathbf{y}, \mathbf{s})$ is on the central path, then the corresponding value of μ satisfies $\mathbf{x}^\top \mathbf{s} = n\mu$. Likewise, given $\mathbf{x}, \mathbf{s} > \mathbf{0}$, define

$$\mu(\mathbf{x}, \mathbf{s}) := \frac{\mathbf{x}^\top \mathbf{s}}{n}.$$

To move from a current point $(\mathbf{x}, \mathbf{y}, \mathbf{s})$ to a new point, we use the so-called *Newton step*; that is, the solution to the system of equations

$$\begin{bmatrix} -\mathbf{Q} & \mathbf{A}^\top & \mathbf{I} \\ \mathbf{A} & \mathbf{0} & \mathbf{0} \\ \mathbf{S} & \mathbf{0} & \mathbf{X} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta \mathbf{y} \\ \Delta \mathbf{s} \end{bmatrix} = \begin{bmatrix} \mathbf{c} + \mathbf{Q}\mathbf{x} - \mathbf{A}^\top \mathbf{y} - \mathbf{s} \\ \mathbf{b} - \mathbf{A}\mathbf{x} \\ \mu \mathbf{1} - \mathbf{X}\mathbf{S}\mathbf{1} \end{bmatrix}. \quad (5.16)$$

Algorithm 5.2 presents a template for an interior-point method.

Algorithm 5.2 Interior-point method for quadratic programming

- 1: choose $\mathbf{x}^0, \mathbf{s}^0 > \mathbf{0}$
 - 2: **for** $k = 0, 1, \dots$ **do**
 - 3: solve the Newton system (5.16) for $(\mathbf{x}, \mathbf{y}, \mathbf{s}) = (\mathbf{x}^k, \mathbf{y}^k, \mathbf{s}^k)$ and $\mu := 0.1\mu(\mathbf{x}^k, \mathbf{s}^k)$
 - 4: choose a step length $\alpha \in (0, 1]$ and set $(\mathbf{x}^{k+1}, \mathbf{y}^{k+1}, \mathbf{s}^{k+1}) = (\mathbf{x}^k, \mathbf{y}^k, \mathbf{s}^k) + \alpha(\Delta \mathbf{x}, \Delta \mathbf{y}, \Delta \mathbf{s})$
 - 5: **end for**
-

The step length α in step 4 should be chosen so that $\mathbf{x}^{k+1}, \mathbf{s}^{k+1} > \mathbf{0}$ and the size of $\mathbf{r}_\mu(\mathbf{x}^{k+1}, \mathbf{y}^{k+1}, \mathbf{s}^{k+1})$ is sufficiently smaller than $\mathbf{r}_\mu(\mathbf{x}^k, \mathbf{y}^k, \mathbf{s}^k)$. A line-search procedure such as the one described in Algorithm 2.4 in Chapter 2 can be used for choosing the step length α .

5.6 Applications to Machine Learning

We next discuss some iconic applications of quadratic programming to machine learning. We must note that the literature on optimization models in machine learning is vast and continues to grow at a rapid pace. For a more detailed discussion on this timely subject, we refer the reader to the excellent textbooks by Friedman et al. (2001), Sra et al. (2012), and Vapnik (2013).

5.6.1 Binary Classification and Support Vector Machines

Classification problems constitute an important class of problems in financial mathematics that can be solved using optimization models and techniques. In a classification problem we have a vector of features describing an entity and the goal is to analyze these features to determine which class each entity belongs to, among two (or more) classes. For example, the classes might be “growth stocks” and “value stocks”, and the entities (stocks) may be described by a feature vector that contains elements such as stock price, price–earnings ratio, growth rate for the previous periods, growth estimates, etc.

Mathematical approaches to classification often start with a training exercise. One is supplied with a list of entities, their feature vectors, and the classes they belong to. From this information, one tries to extract a mathematical structure for the entity classes so that additional entities can be classified using this mathematical structure and their feature vectors. For two-class classification, a hyperplane is probably the simplest mathematical structure that can be used to separate the feature vectors of these two different classes. Of course, there may not be any hyperplane that separates two sets of vectors. When such a hyperplane exists, we say that the two sets can be linearly separated.

Consider feature vectors $\mathbf{a}_i \in \mathbb{R}^n$ for $i = 1, \dots, k_1$ corresponding to class 1, and vectors $\mathbf{b}_i \in \mathbb{R}^n$ for $i = 1, \dots, k_2$ corresponding to class 2. If these two vector sets can be linearly separated, a hyperplane $\mathbf{w}^\top \mathbf{x} = \gamma$ exists with $\mathbf{w} \in \mathbb{R}^n, \gamma \in \mathbb{R}$ such that

$$\begin{aligned}\mathbf{w}^\top \mathbf{a}_i &\geq \gamma, \text{ for } i = 1, \dots, k_1 \\ \mathbf{w}^\top \mathbf{b}_i &\leq \gamma, \text{ for } i = 1, \dots, k_2.\end{aligned}$$

To have a “strict” separation, we often prefer to obtain w and γ such that

$$\begin{aligned}\mathbf{w}^\top \mathbf{a}_i &\geq \gamma + 1, \text{ for } i = 1, \dots, k_1 \\ \mathbf{w}^\top \mathbf{b}_i &\leq \gamma - 1, \text{ for } i = 1, \dots, k_2.\end{aligned}$$

In this manner, we find two parallel lines ($\mathbf{w}^\top \mathbf{x} = \gamma + 1$ and $\mathbf{w}^\top \mathbf{x} = \gamma - 1$) that form the boundaries of the class 1 and class 2 portions of the vector space; see Figure 5.6.

There may be several such parallel lines that separate the two classes. Which one should one choose? A good criterion is to choose the lines that have the

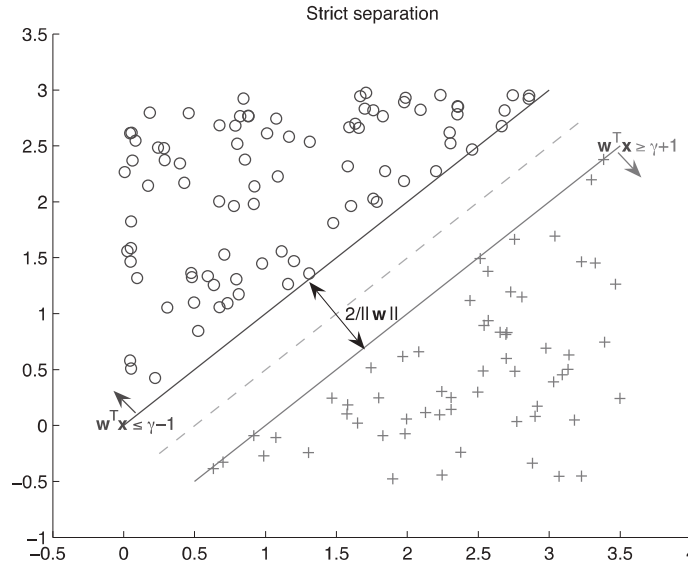


Figure 5.6 Linear separation of two classes of data points

largest margin (distance between the lines). In machine learning, this type of classification model is known as a *support vector machine* (Friedman et al., 2001; Vapnik, 2013).

- (i) Consider the following quadratic problem:

$$\begin{aligned} \min_{\mathbf{w}, \gamma} \quad & \|\mathbf{w}\|_2^2 \\ & \mathbf{a}_i^T \mathbf{w} \geq \gamma + 1, \text{ for } i = 1, \dots, k_1 \\ & \mathbf{b}_i^T \mathbf{w} \leq \gamma - 1, \text{ for } i = 1, \dots, k_2. \end{aligned} \quad (5.17)$$

The objective function of this problem is equivalent to maximizing the margin between the lines $\mathbf{w}^T \mathbf{x} = \gamma + 1$ and $\mathbf{w}^T \mathbf{x} = \gamma - 1$ (see Exercise 5.6).

- (ii) The linear separation idea we presented above can be used even when the two vector sets $\{\mathbf{a}_i\}$ and $\{\mathbf{b}_i\}$ are not linearly separable. (Note that linearly inseparable sets will result in an infeasible problem in formulation (5.17).) This is achieved by introducing a non-negative violation variable for each constraint of (5.17). Then, one has two objectives: to minimize the total of the constraint violations and to maximize the margin. One can formulate a quadratic programming model that combines these two objectives using an adjustable parameter that can be chosen in a way to put more weight on violations or margin, depending on one's preference (see Exercise 5.7).

5.6.2 Ridge and Lasso Regression

Recall the regression problem described in Example 5.3, namely to estimate the linear model

$$Y = \beta^\top X + \epsilon,$$

where X and Y are $\mathbb{R}^p$ -valued and $\mathbb{R}$ -valued random variables, by using some *training data* $(\mathbf{x}_i, y_i)$, with $i = 1, \dots, N$.

We next discuss the case when $N < p$. This case poses a classical and modern challenge in data science. Indeed, this kind of case is increasingly common as modern technology facilitates the collection of data. The expression *high-dimensional problems* in the data science literature (Friedman et al., 2001) is often used to describe problems where $p \gg N$. Examples of high-dimensional problems abound in computational biology and genomics, and other instances will likely emerge. In those contexts N corresponds to the number of individuals, e.g., patients, in some study. Due to physical limitations, N may only be of the order of a few hundred. In contrast, the number of features p that can be gathered, e.g., gene measurements, could be of the order of tens of thousands.

When $p < N$ the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ has rank at most $N < p$ and thus the least-squares approach

$$\min_{\beta} \|\mathbf{X}^\top \beta - \mathbf{y}\|_2^2$$

is inadequate because the optimality conditions lead to an underdetermined system of equations

$$(\mathbf{X}^\top \mathbf{X})\beta = \mathbf{X}^\top \mathbf{y}.$$

We next describe two popular modifications to the ordinary least-squares approach that aim to rectify this difficulty, namely *ridge regression* and *lasso regression*.

Ridge regression adds a quadratic penalty term to the objective function in the least-squares model

$$\min_{\beta} \|\mathbf{X}^\top \beta - \mathbf{y}\|_2^2 + \lambda \|\beta\|_2^2, \quad (5.18)$$

where $\lambda > 0$ is a tuning parameter. The effect of the penalty term is to shrink the regression coefficients towards zero. The magnitude of λ determines the shrinking effect. In the limit when $\lambda \rightarrow \infty$ the solution to the ridge regression model is $\beta = \mathbf{0}$. On the other hand, when $\lambda = 0$ ridge regression and ordinary least squares coincide.

The optimality conditions for (5.18) yield the following system of equations:

$$(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})\beta - \mathbf{X}^\top \mathbf{y} = \mathbf{0}.$$

Thus the solution to (5.18) is

$$\beta = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}.$$

On the other hand, the lasso regression model, proposed in a seminal paper by Tibshirani (1996), adds a 1-norm penalty term to the objective function in the least-squares model

$$\min_{\boldsymbol{\beta}} \|\mathbf{X}^T \boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1, \quad (5.19)$$

where $\lambda > 0$ is a tuning parameter. The effect of the penalty term is again to shrink the regression coefficients towards zero. However, the properties of the 1-norm have a far more interesting effect. The penalty term $\lambda \|\boldsymbol{\beta}\|_1$ makes some of the regression coefficients be *equal to* zero. In particular, the solutions to the lasso regression model (5.19) are typically sparse and the level of sparsity is controlled by the tuning parameter λ . Lasso regression can be formulated as a quadratic program (see Exercise 5.9). Unlike ridge regression, there is no closed-form formula for the solution to lasso regression.

5.7 Exercises

Exercise 5.1 Assume $\mathbf{c} \in \mathbb{R}^n$ and $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is symmetric. Show that the function

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x}$$

is convex if and only if $\mathbf{Q}$ is positive semidefinite.

Assume $\mathbf{c} \in \mathbb{R}^n$ and $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is symmetric and positive semidefinite but not positive definite. Show that the problem

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x}$$

is either bounded or has infinitely many optimal solutions.

Exercise 5.2 Let $\mathbf{c} \in \mathbb{R}^n$. Show that the solution to

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x}\|_2^2 - \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{x} = 1 \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

is

$$\mathbf{x} = (\lambda \mathbf{1} + \mathbf{c})^+,$$

where $\lambda \in \mathbb{R}$ is a suitable threshold value such that $\mathbf{1}^T (\lambda \mathbf{1} + \mathbf{c})^+ = 1$.

Exercise 5.3 Let $\mathbf{c}, \mathbf{d} \in \mathbb{R}^n$. Assume $\mathbf{d} > \mathbf{0}$ and $\mathbf{D} = (\text{Diag}(\mathbf{d}))$. Show that the solution to

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^T \mathbf{D}^{-1} \mathbf{x} - \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{x} = 1 \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

is

$$\mathbf{x} = (\lambda \mathbf{d} + \mathbf{Dc})^+,$$

where $\lambda \in \mathbb{R}$ is a suitable threshold value such that $\mathbf{1}^\top (\lambda \mathbf{d} + \mathbf{Dc})^+ = 1$.

Exercise 5.4 Write a CVX MATLAB script that takes as inputs $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{Q} \in \mathbb{R}^{n \times n}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and solves the optimization problem

$$\begin{aligned} \min \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Test your script on instances generated as follows:

```
>> m=1, n=5, c=randn(n,1), Q=eye(n), A=ones(m,n), b=1;
```

and

```
>> m=1, n=5, c=randn(n,1), Q=diag(rand(n,1)), A=ones(m,n), b=1;
```

Are the results consistent with Exercises 5.2 and 5.3?

Exercise 5.5 Consider a quadratic program with non-negativity inequality constraints only:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{x} \geq \mathbf{0}. \end{aligned} \tag{5.20}$$

There is some intuition behind the optimality conditions: at an optimal solution the non-negativity constraints split into binding and non-binding constraints. The former behave like equality constraints whereas the latter can be treated as if they did not exist. Suppose $I \subseteq \{1, \dots, n\}$ is the set of binding constraints and $J = \{1, \dots, n\} \setminus I$. This reasoning suggests that we think of the problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{x}_I = \mathbf{0}, \end{aligned}$$

whose optimality conditions are

$$\begin{aligned} (\mathbf{Q} \mathbf{x} + \mathbf{c})_I - \mathbf{s}_I &= \mathbf{0} \\ (\mathbf{Q} \mathbf{x} + \mathbf{c})_J &= \mathbf{0} \\ \mathbf{x}_I &= \mathbf{0}. \end{aligned} \tag{5.21}$$

Prove that this intuition is indeed correct.

Exercise 5.6 Consider the quadratic problem (5.17) presented in Section 5.6.1.

- Show that the objective function of this problem is equivalent to maximizing the margin between the lines $\mathbf{w}^\top \mathbf{x} = \gamma + 1$ and $\mathbf{w}^\top \mathbf{x} = \gamma - 1$.
- Write the optimality conditions for problem (5.17).
- Write the dual.

Exercise 5.7 The linear separation idea presented in Section 5.6.1 can be used even when the two vector sets $\{\mathbf{a}_i\}$ and $\{\mathbf{b}_i\}$ are not linearly separable. This is achieved by introducing a non-negative violation variable for each constraint of (5.17). Then, one has two objectives: to minimize the total of the constraint violations and to maximize the margin. Develop a quadratic programming model that combines these two objectives using an adjustable parameter that can be chosen in a way to put more weight on violations or margin, depending on one's preference.

Exercise 5.8 The classification problems discussed in the two previous exercises can also be formulated as linear programming problems, if one agrees to use the 1-norm rather than the 2-norm of $\mathbf{w}$ in the objective function. Recall that $\|\mathbf{w}\|_1 = \sum_i |w_i|$. Show that if we replace $\|\mathbf{w}\|_2^2$ by $\|\mathbf{w}\|_1$ in the objective function of (5.17), we can write the resulting problem as a linear program. Show also that this new objective function is equivalent to maximizing the distance between $\mathbf{w}^\top \mathbf{x} = \gamma + 1$ and $\mathbf{w}^\top \mathbf{x} = \gamma - 1$ if one measures the distance using the ∞ -norm $\|\mathbf{g}\|_\infty = \max_i |g_i|$.

Exercise 5.9 Show that the lasso regression model (5.19) can be equivalently formulated as

$$\begin{array}{ll} \min_{\mathbf{x}} & \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{D} \mathbf{x} \geq \mathbf{d} \end{array}$$

for some suitable $\mathbf{Q}, \mathbf{c}, \mathbf{D}, \mathbf{d}$.